

Lec 6-1. Oct 14, 2024.

- Antiderivative and indefinite integrals (4.7)
- Estimation with finite sums (5.1)
- Riemann sums (5.1)
- Finite sums (5.2)
- Definite integrals : definitions (5.3)
 - Nonintegrable functions
 - Integrable functions

• Indefinite integrals

$$F' = f$$

Def: The set of all antiderivatives of f is called the indefinite integral of f . If F is one antiderivative of f , we write

$$\int f(x) dx = F(x) + C,$$

where C is a constant.

Names :

$$\int \underbrace{f(x)}_{\text{integrand}} \underbrace{dx}_{\text{variable of integration}} \rightarrow \text{indefinite integral of } f \text{ with respect to } x.$$

e.g. $\int \underline{(x^2 - 2x + 5)} dx = \frac{1}{3}x^3 - x^2 + 5x + C$, where C is any constant.

Rmk: $\int [\alpha f(x) + \beta g(x)] dx = \alpha \cdot \int f(x) dx + \beta \cdot \int g(x) dx$.

e.g. $\int (x + x^2) dx = \int x dx + \int x^2 dx$
 $= \frac{1}{2}x^2 + C_1 + \frac{1}{3}x^3 + C_2$
 $= \frac{1}{2}x^2 + \frac{1}{3}x^3 + (C_1 + C_2)$

$= \frac{1}{2}x^2 + \frac{1}{3}x^3 + C,$

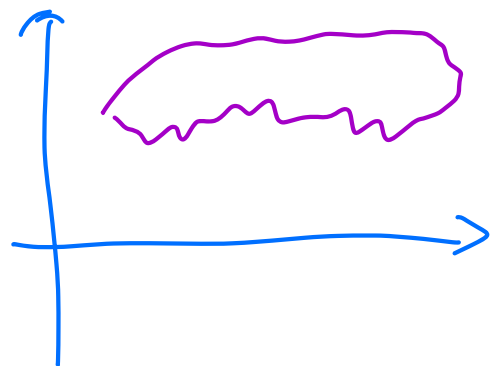
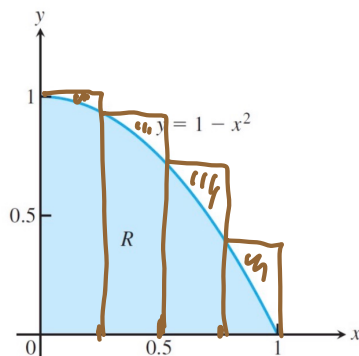
Where we write C instead of $C_1 + C_2$ as this constant is arbitrary. $\square$

	definition	one-sided	infinite	at infinity	laws & Rules	basic func'	Does Not Exist
limit							sandwich thm
continuity							intermediate value thm
derivative					chain rule		
Application	① asymptotes			④ mean value thm		② Newton's method	
	② implicit differentiation			⑤ extreme values		⑦ Antiderivative	
	③ linearization			- monotone: 1-derivative			
				- concavity: 2-derivative			

Chap 5. Integrals

Motivation : Finding Area

Consider finding the area R under the graph of the function $y = 1 - x^2$, above the x -axis, between the vertical lines $x = 0$ and $x = 1$.

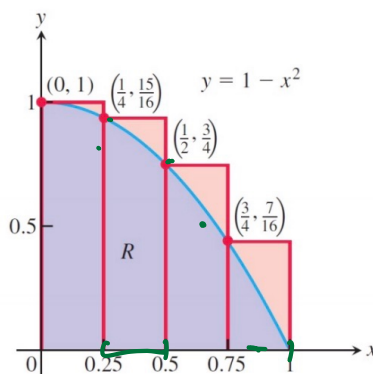
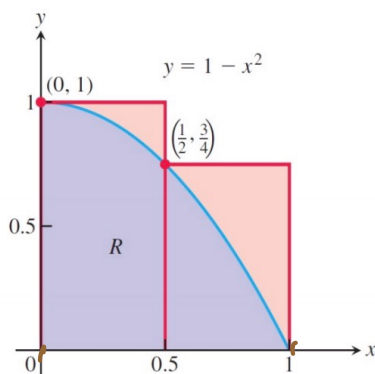


Note that $1 - x^2 \geq 0$ for all $x \in [0, 1]$.

Approximating Area by Rectangles

We may approximate R by summing areas of rectangles:

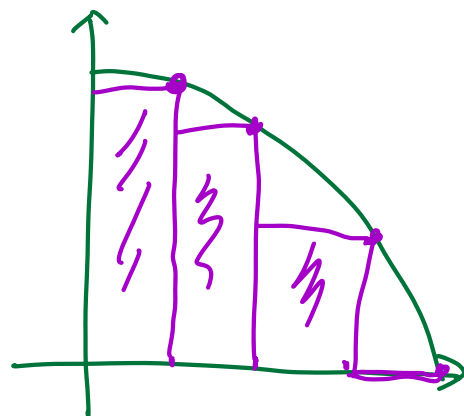
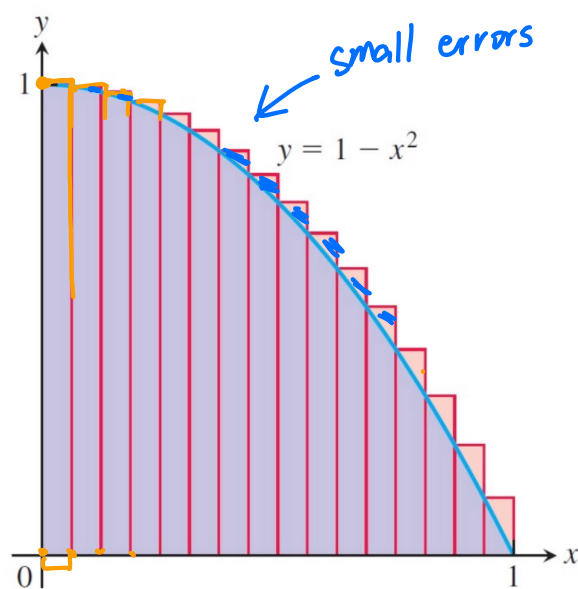
- Divide $[0, 1]$ into subintervals with equal length, and construct rectangles using the function values of the **left endpoints**, as demonstrated in the figure below.
- The sum of the areas of these rectangles is an approximation of R .



Rmk: ① $S = \frac{1}{2} f(0) + \frac{1}{2} f(\frac{1}{2}) = 0.875$; $S = \frac{1}{4} (f(0) + f(\frac{1}{4}) + f(\frac{1}{2}) + f(\frac{3}{4})) = \frac{25}{32} = 0.78125$
 $= \frac{1}{2} \cdot 1 + \frac{1}{2} (1 - (\frac{1}{2})^2)$

②. These approximations are greater than the real area.

Intuitively, this approximation becomes better if we divide $[0, 1]$ into more subintervals.



In the above example :

- ① We divide $[0, 1]$ into n subintervals of length $\Delta x = \frac{1}{n}$, corresponding to points $x_0, x_1, \dots, x_n$, where

$$x_0 = 0, \quad x_1 = \frac{1}{n}, \quad x_2 = \frac{2}{n}, \quad \dots, \quad x_n = \frac{n}{n} = 1.$$

- ② The approximated area is

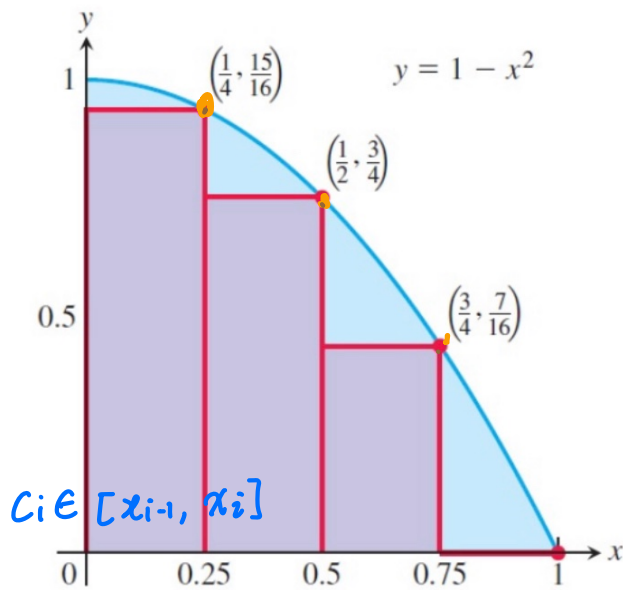
$$\frac{1}{n} \cdot [f(c_1) + f(c_2) + f(c_3) + \dots + f(c_n)] = \sum_{i=1}^n f(c_i) \cdot \frac{1}{n},$$

where each $\underline{c_i} \in \underline{x_{i-1}}, x_i]$.

this notation is called "Sigma".

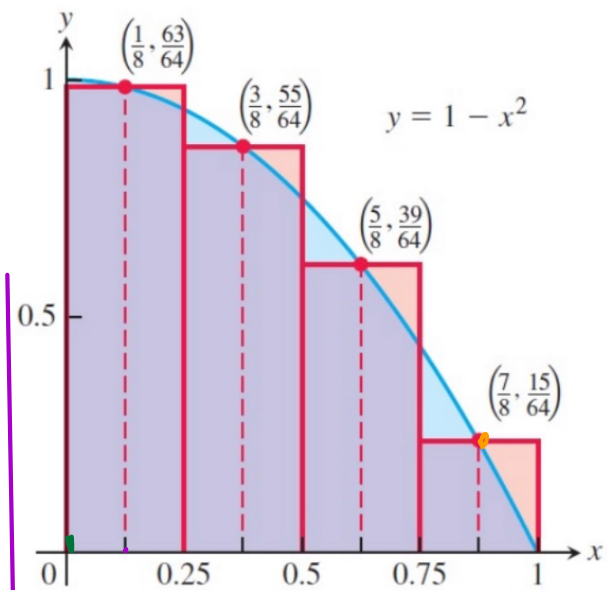
- ③ In this example, f is decreasing, so $c_i = x_{i-1}$ always gives the largest f -value on $[x_{i-1}, x_i]$; when c_i is chosen to give largest (maximum), the sum $\frac{1}{n} \sum_{i=1}^n f(c_i)$ is called an upper sum.

(4) Instead of **left endpoints**, one may also approximate by taking **midpoints** or **right endpoints** of the subintervals:



$$= \frac{1}{4} \cdot (f(\frac{1}{4}) + f(\frac{2}{4}) + f(\frac{3}{4}) + f(\frac{4}{4})) = \frac{1}{4} [f(\frac{1}{8}) + f(\frac{3}{8}) + f(\frac{5}{8}) + f(\frac{7}{8})]$$

$$= \frac{1}{4} \sum_{i=1}^4 f(c_i) \quad (\text{where } c_i = x_i)$$



$$= \frac{1}{4} \sum_{i=1}^4 f(c_i) \quad (\text{where } c_i = \frac{x_{i-1} + x_i}{2})$$

in this
example lower sum

[always smaller than the real area].

TABLE 5.1 Finite approximations for the area of R

Number of subintervals $=n$	Right-endpoint sum	Midpoint sum	Left-endpoint sum
2	0.375	0.6875	$\frac{7}{8} = 0.875$
4	0.53125	0.671875	0.78125
16	0.634765625	0.6669921875	0.697265625
50	0.6566	0.6667	0.6766
100	0.66165	0.666675	0.67165
1000	0.6661665	0.66666675	0.6671665

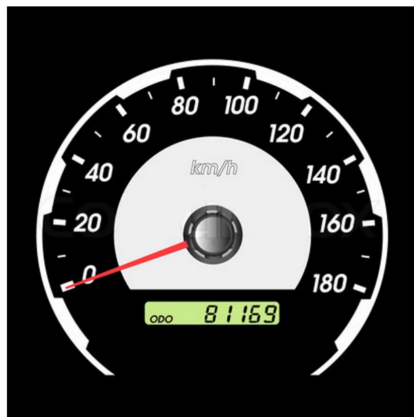
we have ✓
computed above ✓

Q: If $v(t)$ is the velocity of an object and $v(t) \geq 0$ for all $t \in [a, b]$, what is the meaning of the sum

$$\frac{1}{n} (b-a) \sum_{i=1}^n v(x_i) ,$$

where $x_i = a + \frac{b-a}{n} i$, $i = 1, 2, \dots, n$?

Suppose that you are sitting in a car. How can you tell how far you have travelled without checking the signs on the road? Check your speed meter often!



Suppose that the speed of a car is observed every two seconds, and is recorded in the following table.

Time (sec)	0	2	4	6	8	10
Velocity (ft/sec)	20	30	38	44	48	50

If we estimate using the starting speed in each interval, then the total distance travelled is

$$20 \cdot 2 + 30 \cdot 2 + 38 \cdot 2 + 44 \cdot 2 + 48 \cdot 2 = 360 \text{ feet.}$$

If we estimate using the ending speed in each interval, then the total distance travelled is

$$30 \cdot 2 + 38 \cdot 2 + 44 \cdot 2 + 48 \cdot 2 + 50 \cdot 2 = 420 \text{ feet.}$$

How if the speed is observed every second?

Time (sec)	0	1	2	3	4	5	6	7	8	9	10
Velocity (ft/sec)	20	26	30	35	38	42	44	46	48	49	50

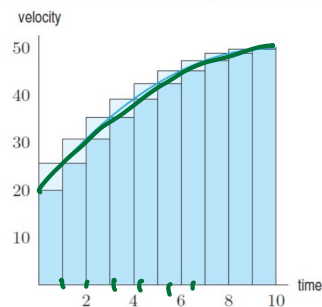
Again using the starting speed and ending speed to estimate, the total distance travelled is approximately:

$$20 + 26 + 30 + 35 + 38 + 42 + 44 + 46 + 48 + 49 = 378 \text{ feet}$$

and

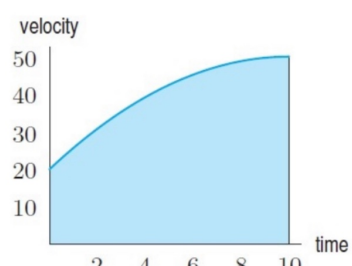
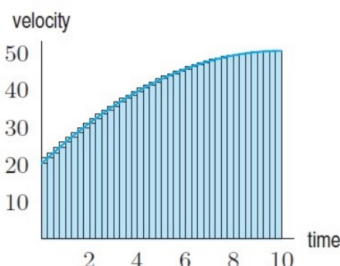
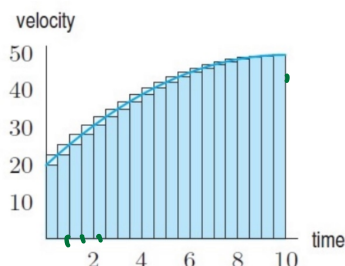
$$26 + 30 + 35 + 38 + 42 + 44 + 46 + 48 + 49 + 50 = 408 \text{ feet.}$$

Time (sec)	0	1	2	3	4	5	6	7	8	9	10
Velocity (ft/sec)	20	26	30	35	38	42	44	46	48	49	50



Dark blue = using starting speed; light blue = using ending speed.

*Area = total
distance
travelled.*



Riemann Sums

Def: A ^① partition of the interval $[a, b]$ is a set $P = \{x_0, x_1, \dots, x_n\}$

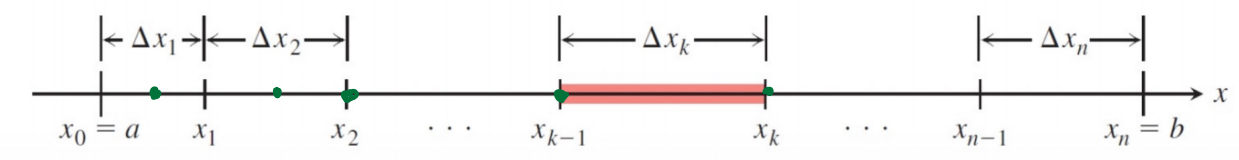
s.t. $a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$.

Given a function $f: [a, b] \rightarrow \mathbb{R}$ with a partition P of $[a, b]$,

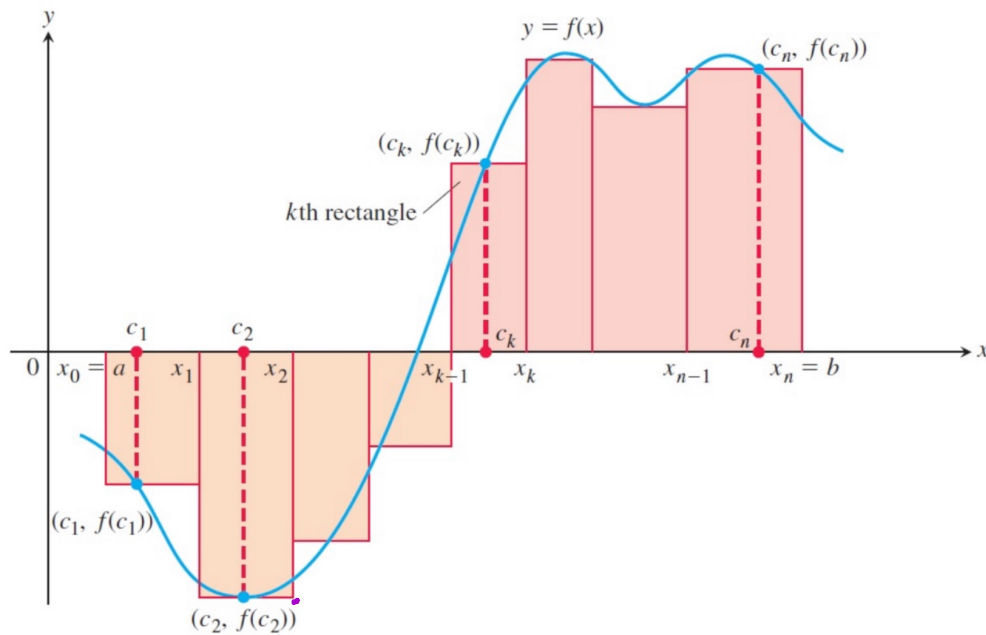
a Riemann sum of f with respect to P is a sum of the form

$$\begin{aligned} S_P &= \sum_{k=1}^n f(c_k) \cdot (x_k - x_{k-1}) \\ &= \sum_{k=1}^n f(c_k) \cdot \Delta x_k \end{aligned}$$

where ^② $c_k \in [x_{k-1}, x_k]$ and ^③ $\Delta x_k = x_k - x_{k-1}$ for each $k \in \{1, \dots, n\}$.



Note that there are many Riemann sums for a function f : a Riemann sum depends on the partition P and the points c_k chosen from the subintervals.



- The left-hand, right-hand, and midpoint sums are all special cases of a Riemann sum.
- In general, Δx_k may be different for different k .

Finite sums

— Sigma notation : $\sum_{i=1}^n a_i \triangleq \sum_{i \in \{1, 2, \dots, n\}} a_i = a_1 + \dots + a_n$

— Linearity of Σ : $\sum_{i=1}^n (ka_i + tb_i) = k \sum_{i=1}^n a_i + t \sum_{i=1}^n b_i$

$\hookrightarrow (ka_1 + tb_1) + (ka_2 + tb_2) + \dots + (ka_n + tb_n) \quad \checkmark$

$= k(a_1 + \dots + a_n) + t(b_1 + b_2 + \dots + b_n).$

$$- \sum_{i=1}^n a_i = \sum_{j=1}^n a_j$$

↙ ↘

The choice of symbols are not important.

Such a variable i or j is called a dummy variable.

The following are some useful identities:

$$\blacktriangleright \sum_{k=1}^n k = \overset{n + (n-1) + \dots + 1}{1 + 2 + \dots + n} = \frac{1}{2}n(n+1). \quad (1)$$

$$\blacktriangleright \sum_{k=1}^n k^2 = 1^2 + 2^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1). \quad (2)$$

$$\blacktriangleright \sum_{k=1}^n k^3 = 1^3 + 2^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2. \quad (3)$$

(1) is the famous identity observed by 7-year-old Gauss.

Q: Is there a method to find $\sum_{k=1}^n k^m$ for a fixed m ?

A: Start with n^{m+1} and use formulae derived for a smaller m .

e.g. Find $\sum_{k=1}^n k^2$.

Sol: Step 1: Start with n^3 !

$$\text{Let } S_n = \sum_{k=1}^n k^2.$$

$$\text{Note that } n^3 = n^3 - (n-1)^3 + (n-1)^3 - (n-2)^3 + \dots + 1^3 - 0^3$$

$$= \sum_{k=1}^n [k^3 - (k-1)^3]$$

$$= \sum_{k=1}^n [k^3 - (k-1)(k^2 - 2k + 1)]$$

$$= \sum_{k=1}^n [k^3 - (k^3 - 3k^2 + 3k - 1)]$$

$$= \sum_{k=1}^n [3k^2 - 3k + 1]$$

$$= 3 \sum_{k=1}^n k^2 - 3 \sum_{k=1}^n k + n$$

Step 2. Solve S_n .

$$S_n = \frac{1}{3} \left(n^3 - n + 3 \sum_{k=1}^n k \right)$$

$$= \frac{1}{3} \left(n^3 - n + 3 \frac{n(n+1)}{2} \right)$$

$$= \frac{1}{3} (n^3 - n) + \frac{1}{2} n(n+1)$$

$$= \frac{1}{6} n(n+1)(2n+1).$$

e.g. Prove the formula ③ by yourself.

e.g. Use right-hand sum approximation and limits to determine the shaded area.

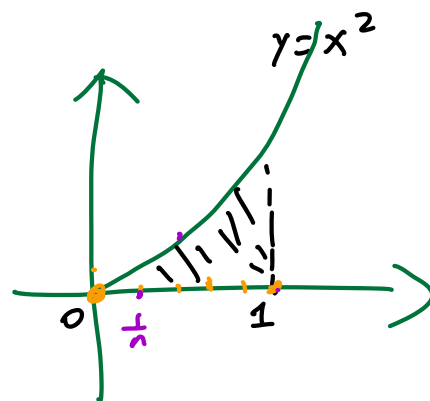
$$\text{Sol}': S_n = \sum_{k=1}^n \frac{1}{n} f\left(\frac{k}{n}\right)$$

$$= \sum_{k=1}^n \frac{1}{n} \left(\frac{k}{n}\right)^2$$

$$= \frac{1}{n^3} \sum_{k=1}^n k^2$$

$$= \frac{1}{n^3} \cdot \frac{1}{6} n(n+1)(2n+1)$$

$$= \frac{(n+1)(2n+1)}{6n^2}$$



Letting $n \rightarrow \infty$, we have $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3}$,

which is the approximation of the shaded area.

□.

$$Q: \int_0^1 x^2 dx = ?$$

Definite Integrals (Riemann integrals)

Definition

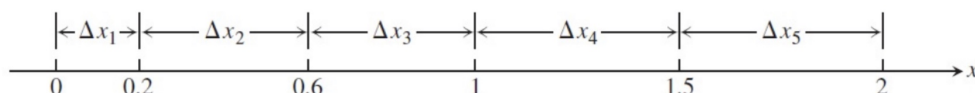
Let $P := \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. The **norm** of P , denoted by $\|P\|$, is defined by

$$\|P\| := \max_{k: 1 \leq k \leq n} \Delta x_k.$$

That is, $\|P\|$ is the length of the largest subinterval given by P .

Example

The partition P of $[0, 2]$ represented by the following figure has norm $\|P\| = 0.5$.



$$P = \{0, 0.2, 0.6, 1, 1.5, 2\} \Rightarrow \|P\| = 0.5.$$

DEFINITION Let $f(x)$ be a function defined on a closed interval $[a, b]$. We say that a number J is the **definite integral of f over $[a, b]$** and that J is the limit of the Riemann sums $\sum_{k=1}^n f(c_k) \Delta x_k$ if the following condition is satisfied:

Given any number $\epsilon > 0$ there is a corresponding number $\delta > 0$ such that for every partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of c_k in $[x_{k-1}, x_k]$, we have

$$\left| \sum_{k=1}^n f(c_k) \Delta x_k - J \right| < \epsilon.$$

In plain English:

J is the limit of the Riemann sums of f

if all Riemann sums of f are arbitrarily close to J for

all sufficiently fine partitions.

δ small = max Δx_k small = fine

rough



Rmk : With the definition above :

• We write $\lim_{\|P\| \rightarrow 0} \underbrace{\sum_{k=1}^n f(c_k) \Delta x_k}_{\text{Riemann sums}} = J$.

- If the limit J exists, we say that f is integrable on $[a, b]$, and write the limit as $\int_a^b f(x) dx$ (i.e., able to be integrated finitely)

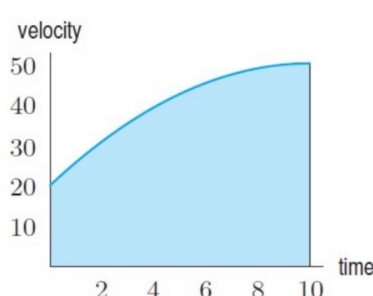
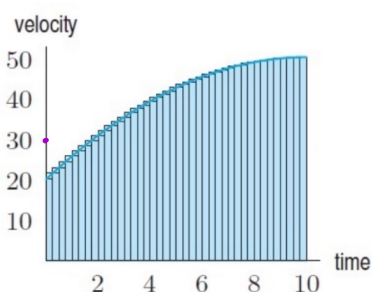
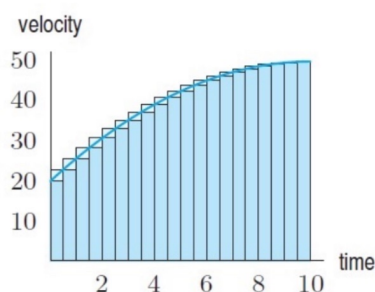
This symbol is called the definite integral or the Riemann integral of f on $[a, b]$ (or from a to b).

DEFINITION Let $f(x)$ be a function defined on a closed interval $[a, b]$. We say that a number J is the **definite integral of f over $[a, b]$** and that J is the limit of the Riemann sums $\sum_{k=1}^n f(c_k) \Delta x_k$ if the following condition is satisfied:

③ Given any number $\epsilon > 0$ there is a corresponding number $\delta > 0$ such that for every partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of c_k in $[x_{k-1}, x_k]$, we have

$$\left| \sum_{k=1}^n f(c_k) \Delta x_k - J \right| < \epsilon.$$

$$J = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \cdot \Delta x_k = \int_a^b f(x) dx \Rightarrow f \text{ is integrable on } [a, b].$$



- Definite integrals (Riemann integrals)

goal :

① Which function is integrable ?

② If f is integrable, then how to compute $\int_a^b f(x) dx$?

Given f defined on $[a, b]$ and a partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$, define

$$M_k \triangleq \max_{x \in [x_{k-1}, x_k]} f(x)$$

and

$$m_k \triangleq \min_{x \in [x_{k-1}, x_k]} f(x), \text{ for } k = 1, 2, \dots, n.$$

Then $U_P(f) \triangleq \sum_{k=1}^n M_k \cdot \Delta x_k$ is an upper sum of f

and $L_P(f) \triangleq \sum_{k=1}^n m_k \cdot \Delta x_k$ is a lower sum of f .

Observe that $U_P(f)$ can only decrease as P gets finer ($\|P\|$ gets smaller)
and $L_P(f)$ can only increase as P gets finer.

Intuition to show this :

Let $P_1 = \{x_0, x_1, \dots, x_n\}$ and $\max_{k=1, \dots, n} \Delta x_k = \Delta x_j = x_j - x_{j-1}$

Let $P_2 = \{x_0, x_1, \dots, x_{j-1}, y, x_j, x_{j+1}, \dots, x_n\}$

Hence, $\|P_2\| \leq x_j - x_{j-1} = \Delta x_j = \|P_1\|$

Compute :

$$U_{P_2}(f) = \sum_{k=1}^{j-1} M_k \Delta x_k + \max_{x \in [x_{j-1}, y]} f(x) \cdot (y - x_{j-1})$$

$$+ \sum_{k=j+1}^n M_k \Delta x_k + \max_{x \in [y, x_j]} f(x) \cdot (x_j - y)$$

$$\leq \sum_{k=1}^{j-1} M_k \Delta x_k + \max_{x \in [x_{j-1}, x_j]} f(x) \cdot (y - x_{j-1})$$

$$+ \sum_{k=j+1}^n M_k \Delta x_k + \max_{x \in [x_{j-1}, x_j]} f(x) \cdot (x_j - y)$$

$$= \sum_{k=1}^n M_k \cdot \Delta x_k$$

$$= U_{P_1}(f)$$

$$\max_{x \in A} f(x) \leq \max_{x \in B} f(x)$$

if $A \subseteq B$.

✓

$\therefore U_P(f)$ decreases as P gets finer.



Intuition. A function is integrable (on $[a, b]$) if $U_P(f)$ and $L_P(f)$ are finite, and the gap between $U_P(f)$ and $L_P(f)$ goes to 0 as $\|P\| \rightarrow 0$. The reason is that all the Riemann sums would be squeezed by $L_P(f)$ and $U_P(f)$ as $\|P\| \rightarrow 0$, and the squeezed number is the limit J .

$$L_P(f) \leq \sum_{k=1}^n f(c_k) \Delta x_k \leq U_P(f)$$

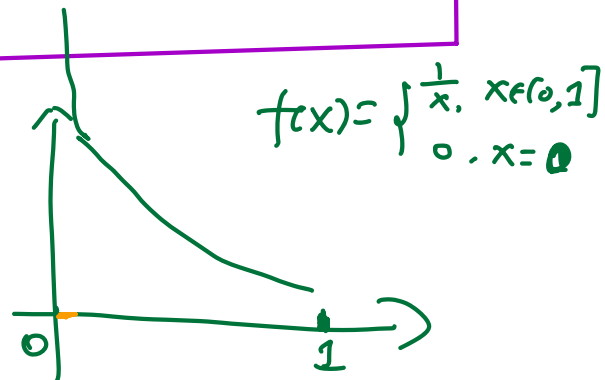
$\|P\| \rightarrow 0 \downarrow$

$\downarrow$

$$J \leq J \leq J \triangleq \int_a^b f(x) dx.$$

Q: which function is integrable?

Thm A. If f is unbounded on $[a, b]$, then it is not integrable on $[a, b]$.



Proof intuition

① • Suppose f is not bounded from above.

[You can do similarly if f is not bounded from below.]

• For any $M > 0$, $\exists x_m \in [a, b]$, s.t. $f(x_m) > M$.

• Take $f(x) = \begin{cases} \frac{1}{x}, & x \in (0, 1] \\ 0, & x = 0 \end{cases}$ as an example.

• For any partition P of $[0, 1]$, consider a Riemann sum

$$\sum_{k=1}^n f(c_k) \Delta x_k.$$

② • Since f is unbounded from above on $[x_0, x_1]$, one may

let $f(c_1) \Delta x_k$

be arbitrarily big by choosing c_1 .

[Note that Δx_1 is small but fixed at the beginning.]

• Keeping all other c_k 's unchanged, we see that there is an arbitrarily big Riemann sum.

• Since this is true for any partition (no matter how small $\|P\|$ is), the Riemann sums have no limit, so f is not integrable on $[0, 1]$.

Pf [Optional]

• Suppose f is not bounded from above.

[You can do similarly if f is not bounded from below.]

• To show: $\forall M > 0, \forall \delta > 0,$

$$\exists \text{ a partition } P, \text{ s.t. } \sum_{k=1}^n f(c_k) \Delta x_k > M.$$

(*)

• We consider the equal partition:

$$a = x_0 < x_1 < \dots < x_n = b \text{ where } x_i = a + \frac{i}{n}(b-a).$$

• As f is unbounded from above on $[a, b]$,

f is unbounded at least on one sub-interval, say, $[x_{j-1}, x_j]$.

• For any subintervals $k=1, \dots, j-1$, we take the left endpoint; $c_k = x_{k-1}$.

For any subintervals $k=j+1, \dots, n$, we take the right endpoint: $c_k = x_k$.

$$\begin{aligned} \therefore \sum_{k=1}^n f(c_k) \Delta x_k &= \sum_{k=1}^{j-1} f(x_{k-1}) \Delta x_k + f(c_j) \Delta x_j \\ &\quad + \sum_{k=j+1}^n f(x_k) \Delta x_k \\ &= \underbrace{\sum_{k=1}^{j-1} f(x_{k-1}) \frac{b-a}{n} + \sum_{k=j+1}^n f(x_k) \frac{b-a}{n} + f(c_j) \frac{b-a}{n}} \end{aligned}$$

• For this fixed M , we find a $c_j \in [x_{j-1}, x_j]$ s.t. $f(c_j) > \frac{n}{b-a} \left(M - \left(\sum_{k=1}^{j-1} f(x_{k-1}) \frac{b-a}{n} + \sum_{k=j+1}^n f(x_k) \frac{b-a}{n} \right) \right)$

• This partition P and $\{c_k\}_{k=1}^n$ provided above induce

$$\text{a Riemann sum } \sum_{k=1}^n f(c_k) \Delta x_k = \sum_{k=1}^{j-1} f(x_{k-1}) \frac{b-a}{n} + \sum_{k=j+1}^n f(x_k) \frac{b-a}{n} + f(c_j) \frac{b-a}{n} > M \text{ for any } M > 0 \text{ and } \delta > 0. \quad \square$$

Rmk. When we talk about definite integrals on $[a, b]$, we may assume that f is bounded on $[a, b]$.